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(54) **SEMICONDUCTOR ELEMENT AND SEMICONDUCTOR DEVICE**

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(57) **ABSTRACT**

A semiconductor element includes a main body with a semiconductor layer, an electrode located on one side of the main body in a first direction and electrically conducting to the semiconductor layer, a terminal opposite to the main body with respect to the electrode in the first direction and electrically conducting to the electrode, and a metal bonding layer electrically conducting to the terminal. The terminal includes a first surface facing a side where the electrode is located in the first direction, and an opposite second surface. The bonding layer is opposite to the first surface across the second surface. In a second direction orthogonal to the first direction, the dimension of the periphery of the first surface is not less than 40% and not greater than 60% of the dimension of the periphery of the second surface.

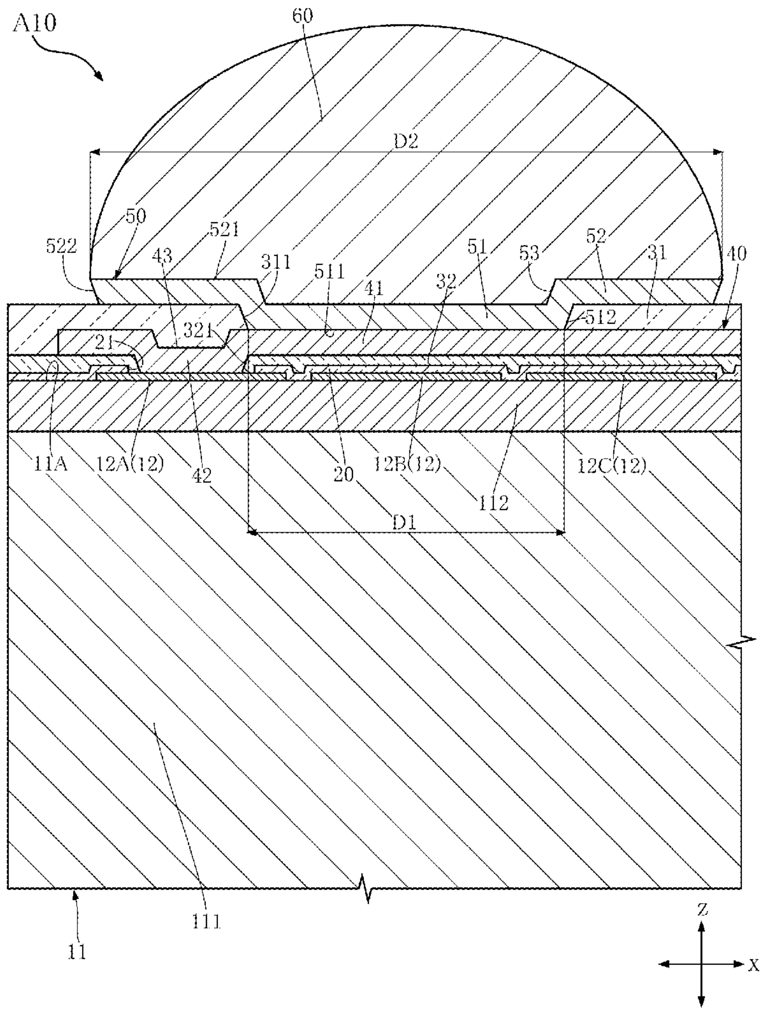


FIG.1

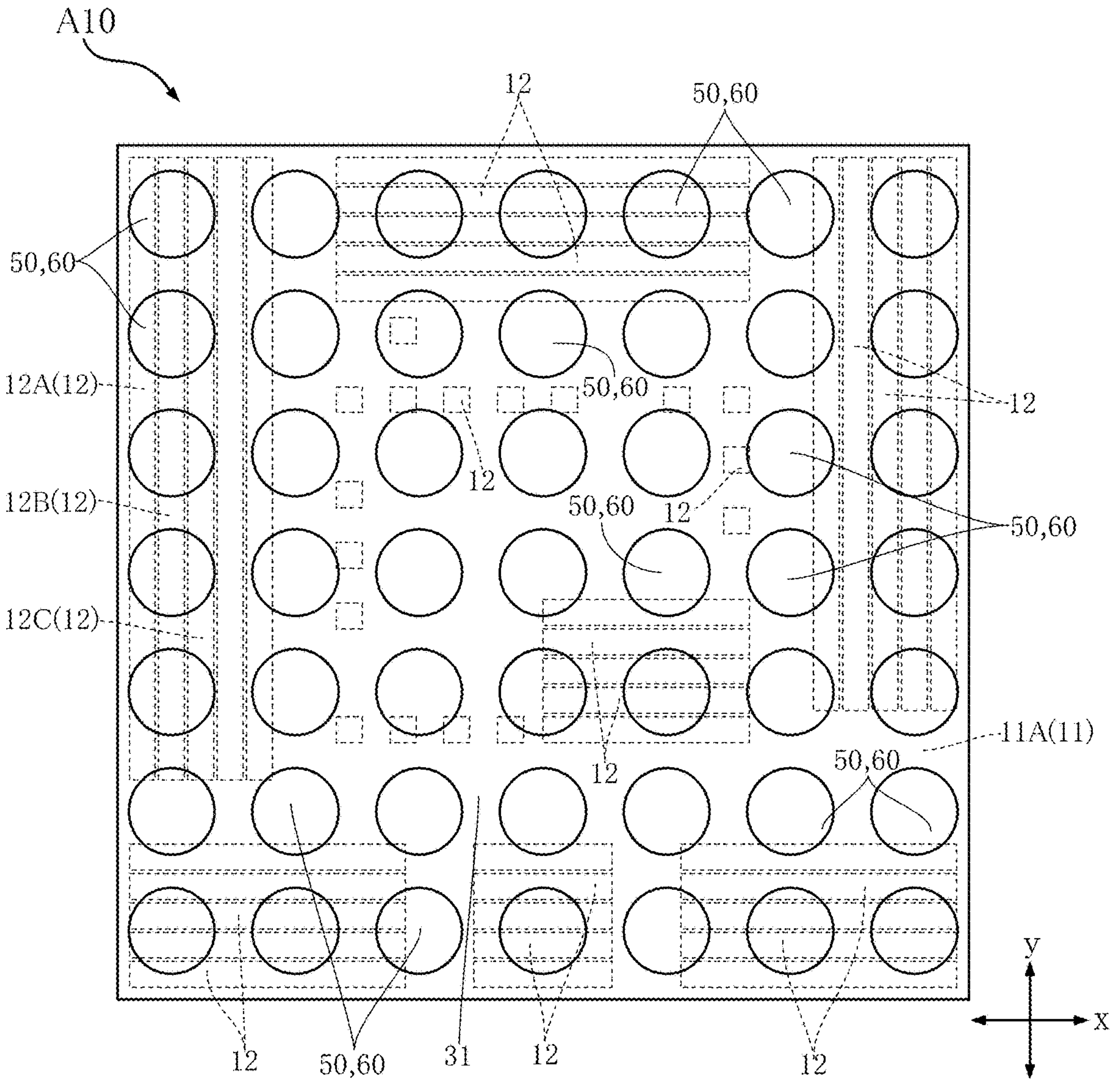
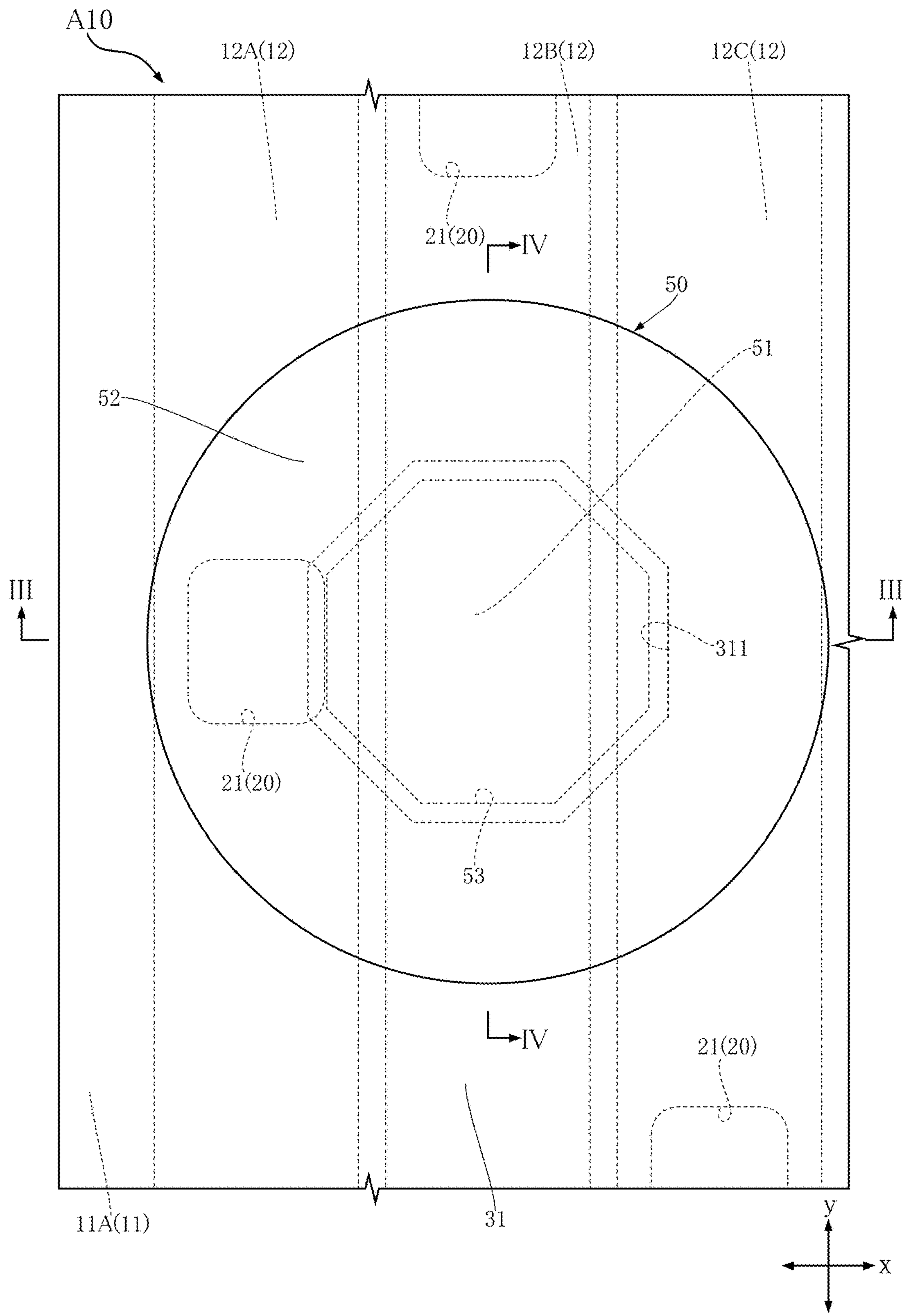

A10
12
50,60
50,60
12A(12)
12B(12)
12C(12)
50,60
12
12
12
50,60
50,60
11A(11)
50,60
12
y
x
12
50,60
31
12
12

FIG.2



A10
12A(12)
12B(12)
12C(12)
21(20)
IV
50
51
52
III
III
311
21(20)
53
IV
21(20)
11A(11)
31
y
x

FIG.3

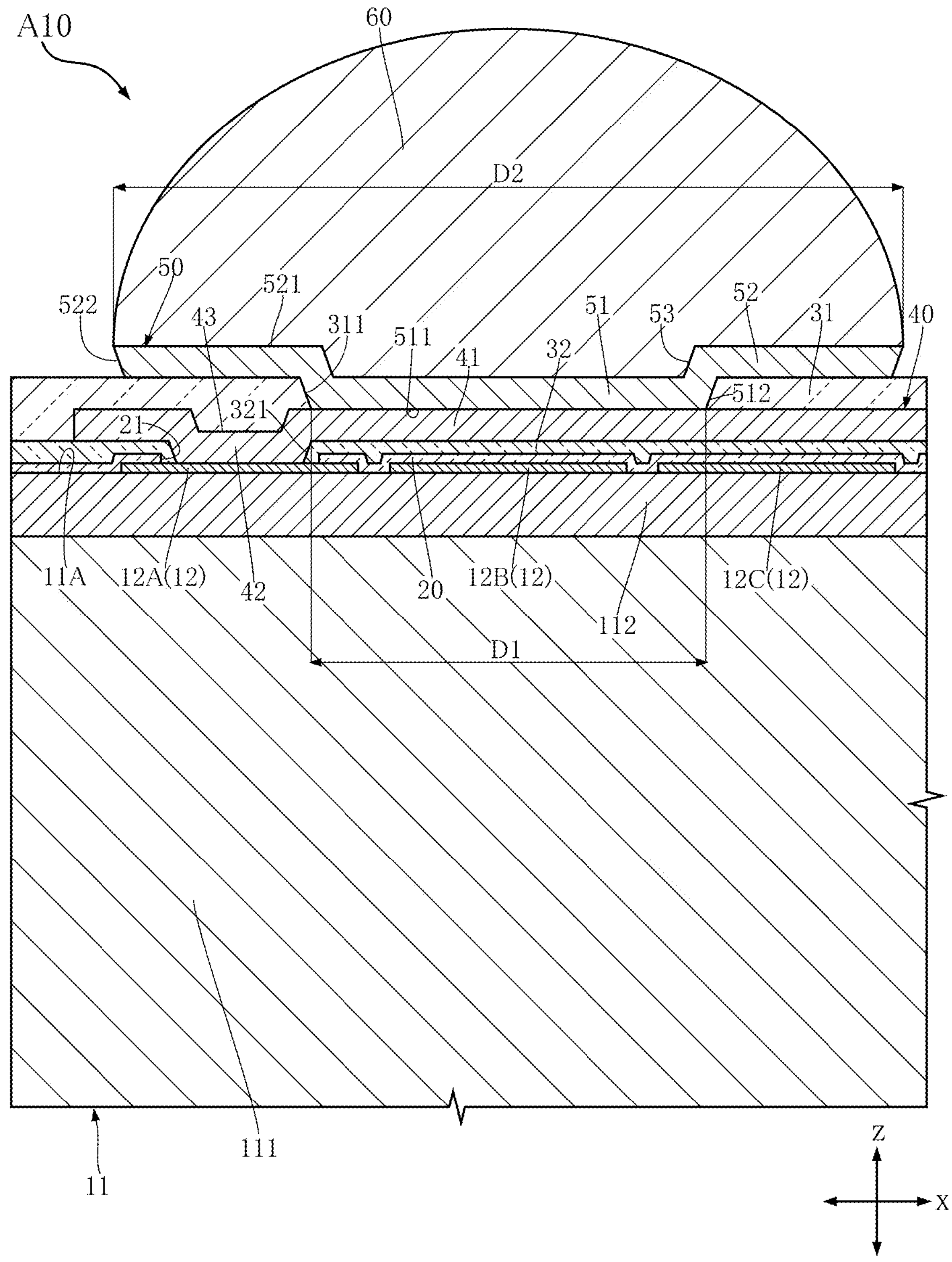

A10
60
D2
50
521
522
43
311
511
41
32
51
53
52
31
40
512
321
21
11A
12A(12)
42
20
12B(12)
112
12C(12)
D1
111
11
Z
X

FIG.4

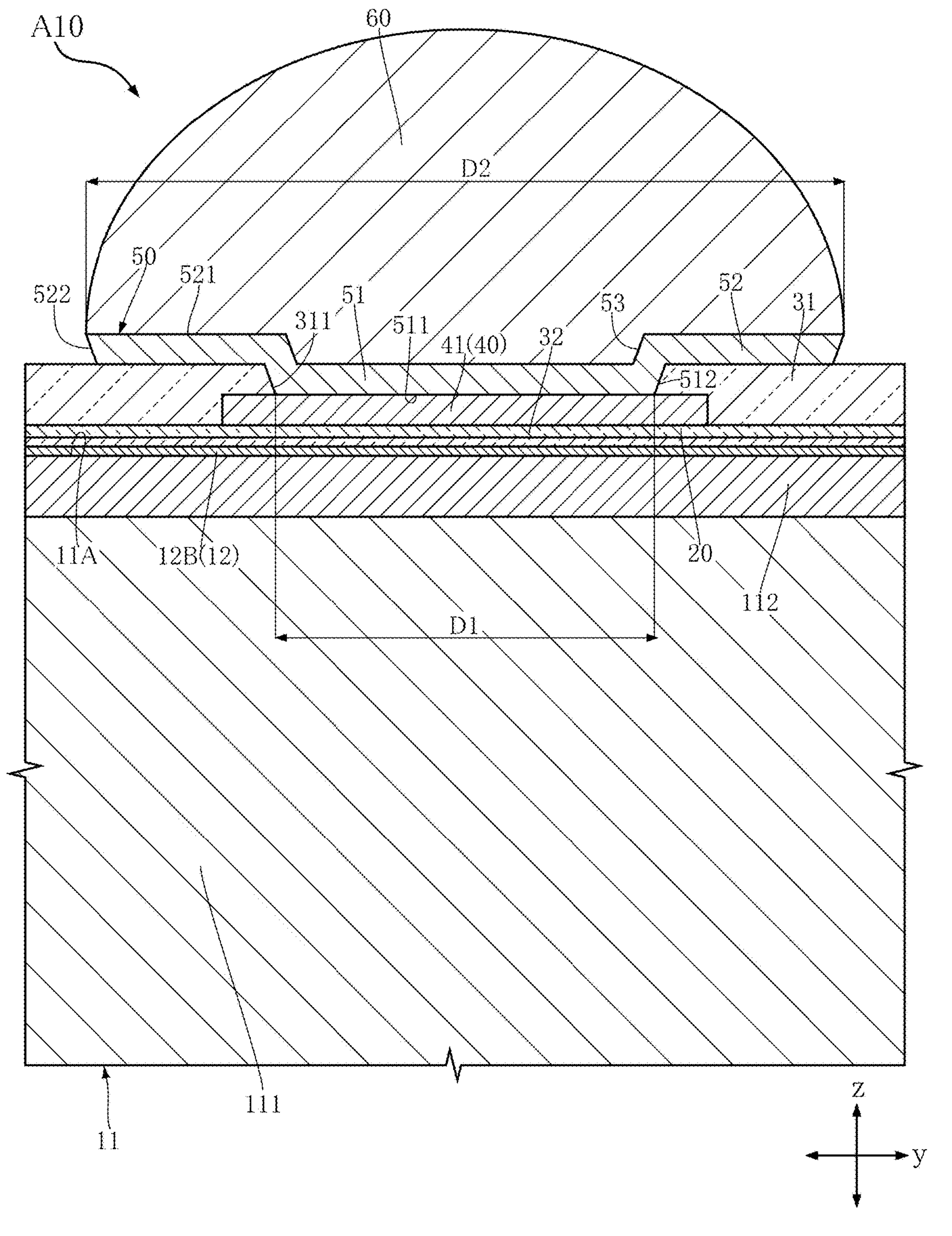

A10
60
D2
50
522
521
51
311
511
41(40)
32
53
52
31
512
11A
12B(12)
20
112
D1
111
11
z
y

FIG.5

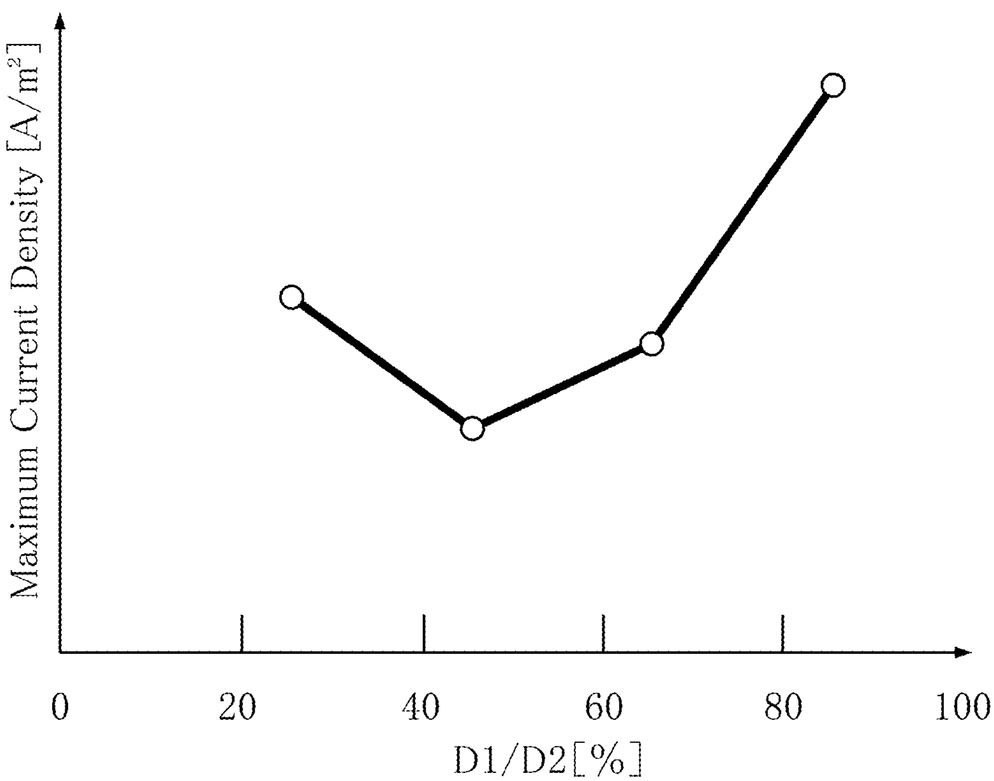


FIG.6

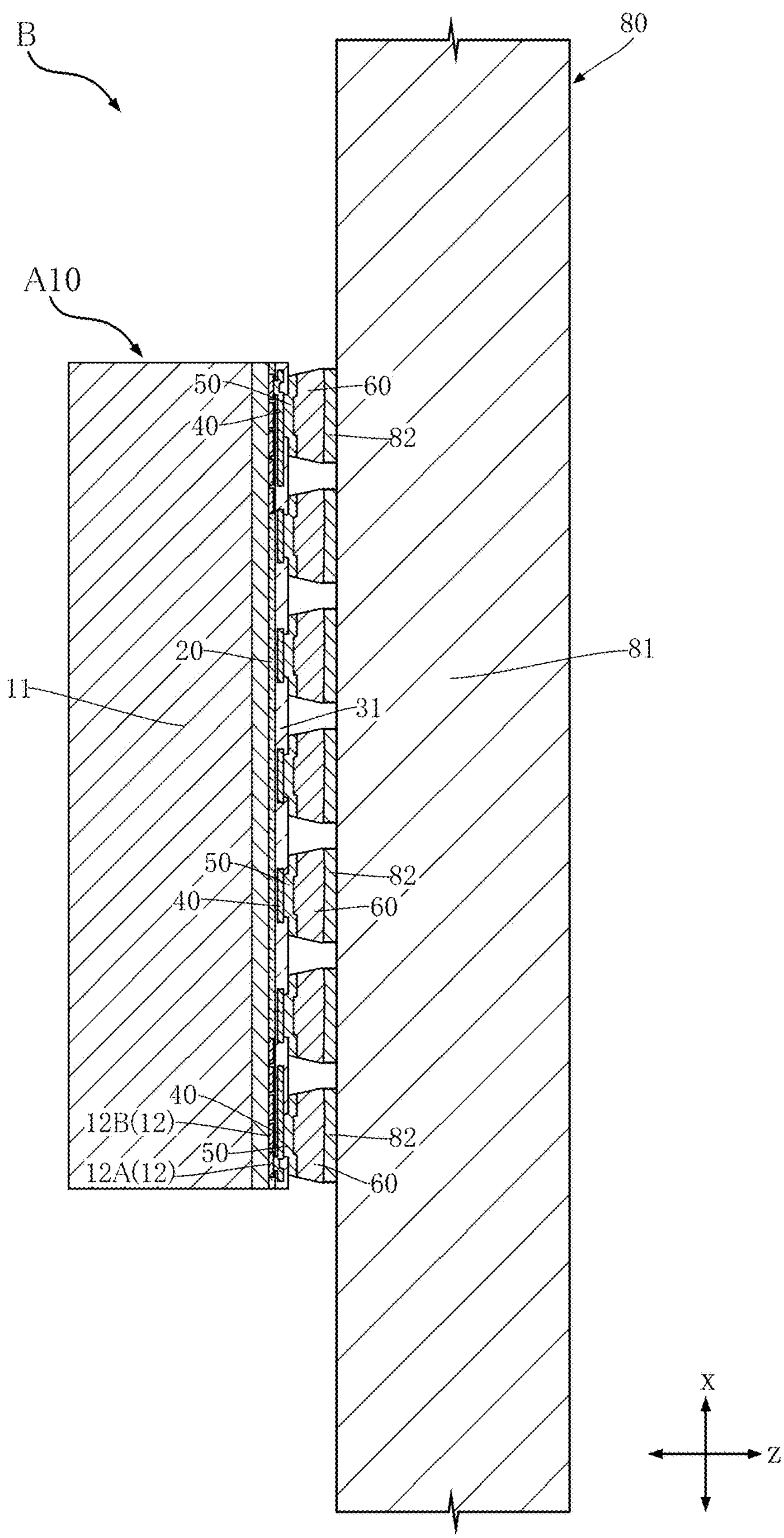

B
80
A10
50
60
40
82
20
81
11
31
50
82
40
60
40
12B(12)
82
50
12A(12)
60
X
Z

FIG.7

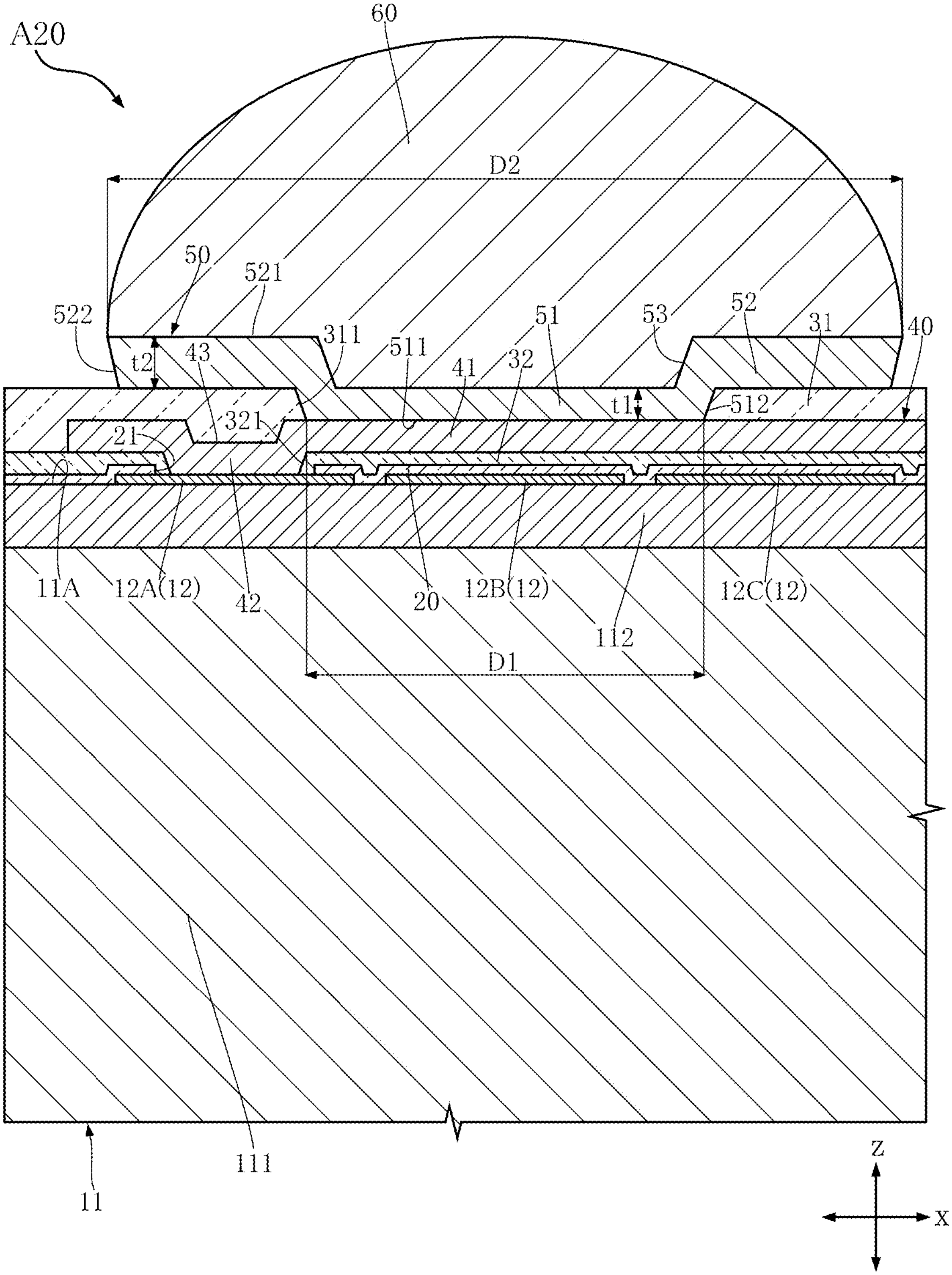

A20
60
D2
50
521
522
t2
43
311
511
41
32
51
53
52
31
40
t1
512
21
321
11A
12A(12)
42
20
12B(12)
112
12C(12)
D1
111
11
Z
X

FIG.8

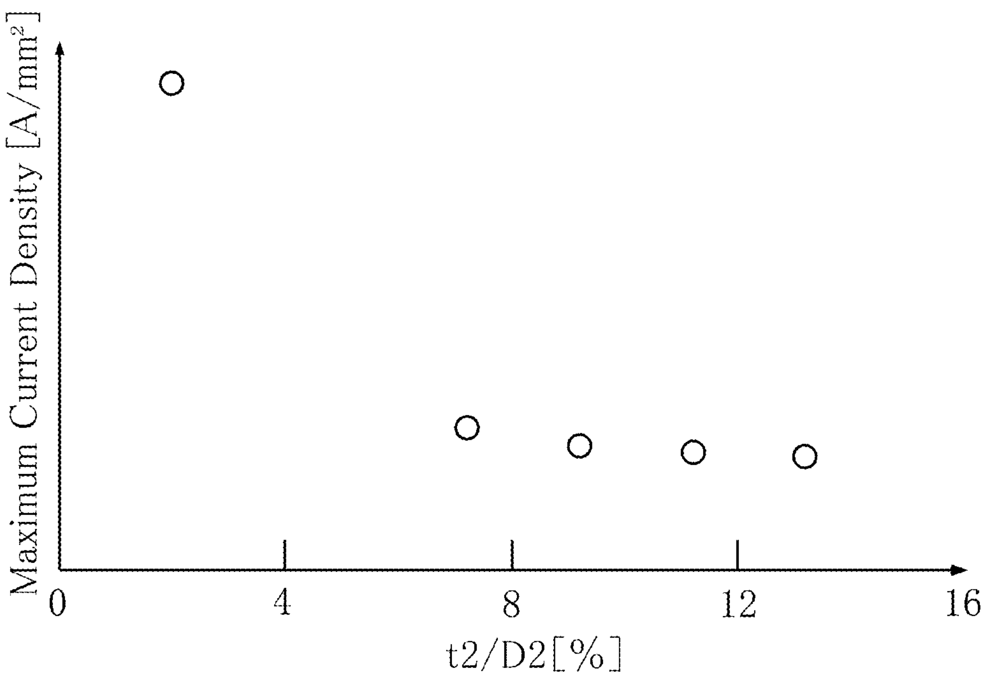


FIG.9

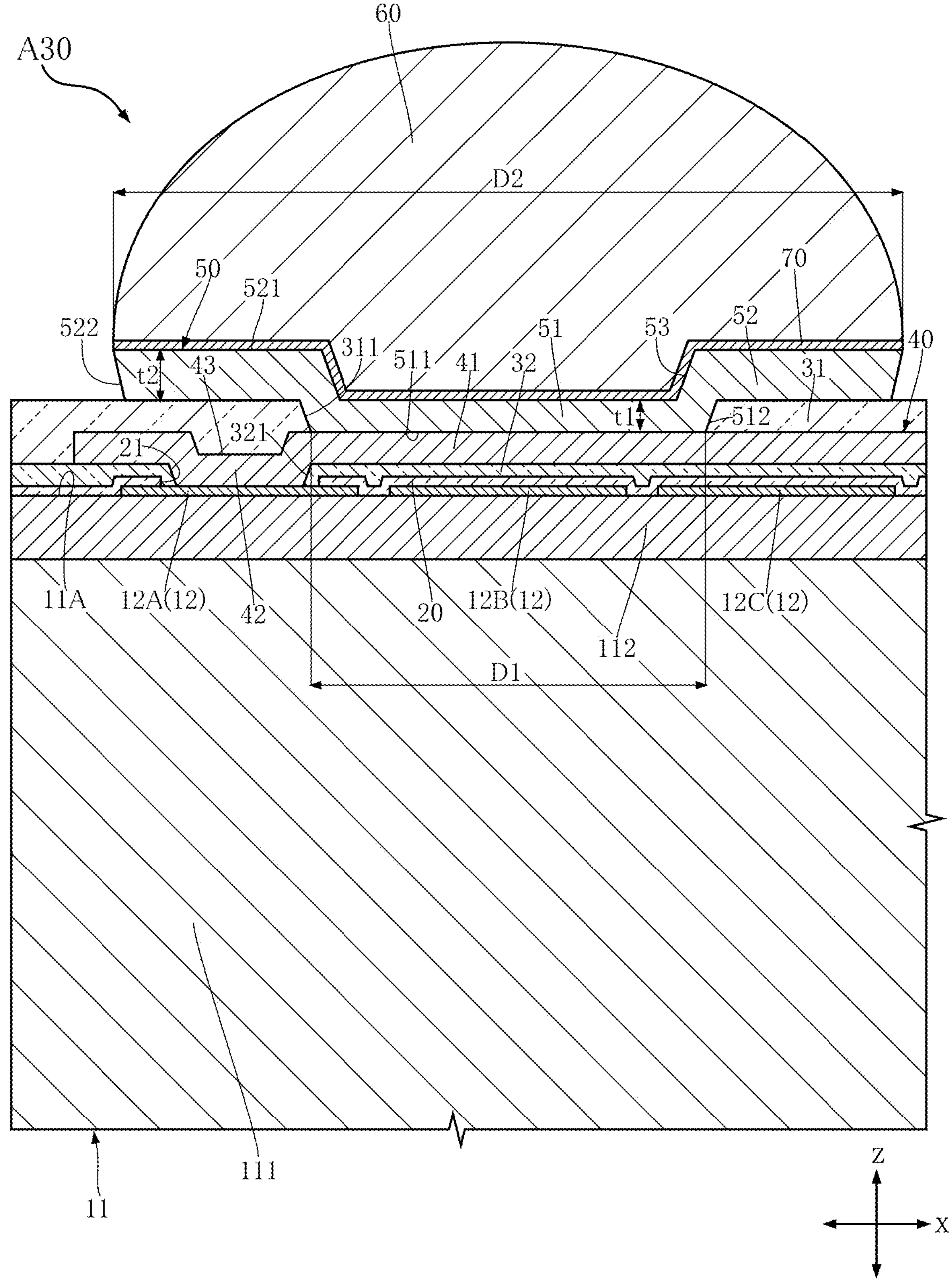

A30
60
D2
50
521
522
t2
43
311
511
41
32
51
53
52
70
40
31
t1
512
321
21
11A
12A(12)
42
20
12B(12)
112
12C(12)
D1
111
11
Z
X

FIG.10

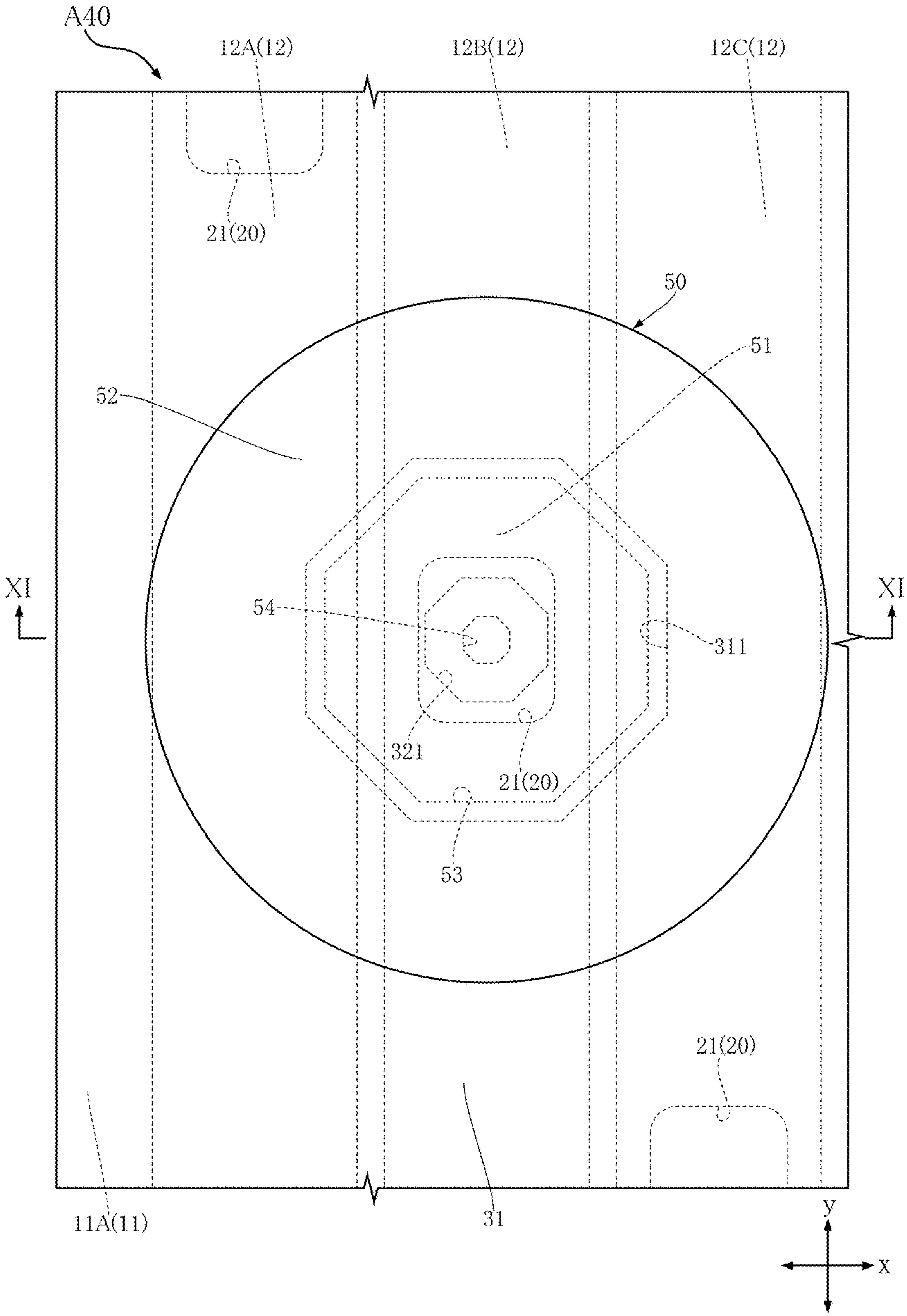

A40
12A(12)
12B(12)
12C(12)
21(20)
50
51
52
XI
54
311
XI
321
21(20)
53
21(20)
11A(11)
31
y
x

FIG.11

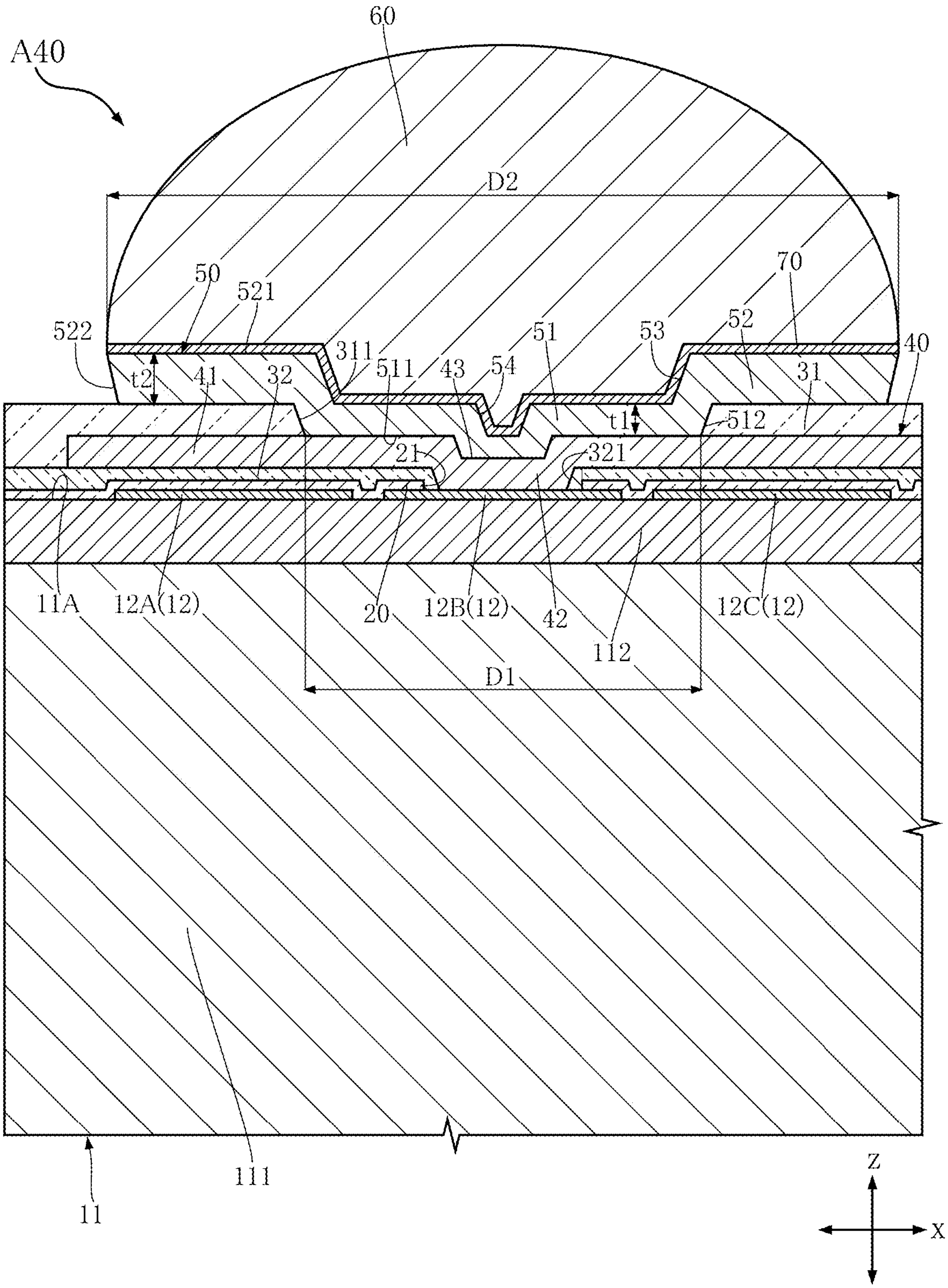
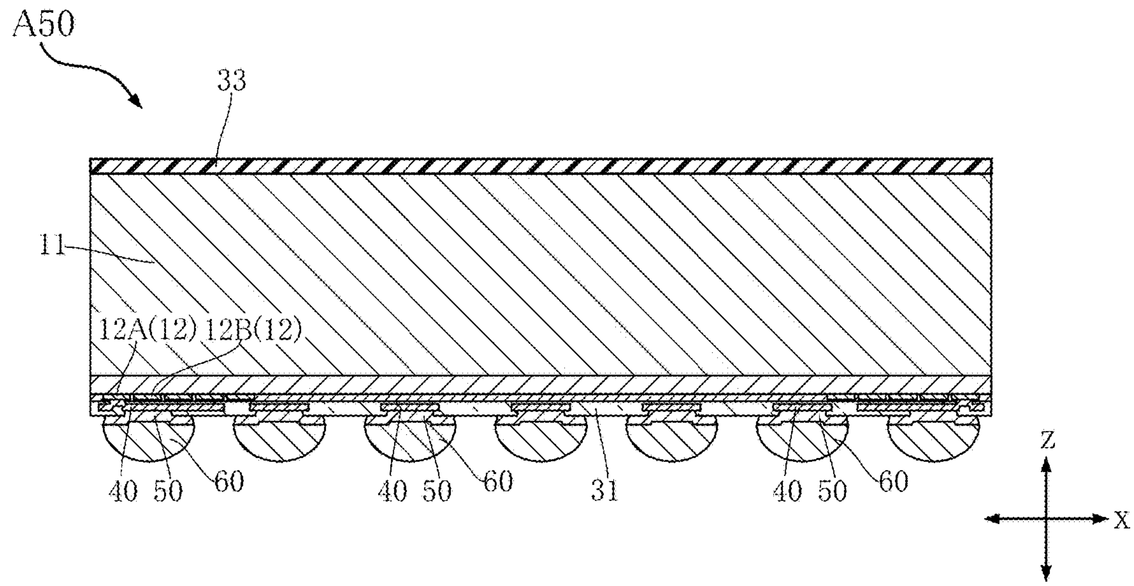

A40
60
D2
50
521
522
t2
41
32
311
511
43
54
51
53
52
70
40
31
t1
512
21
321
11A
12A(12)
20
12B(12)
42
12C(12)
112
D1
111
11
z
x

FIG.12



SEMICONDUCTOR ELEMENT AND SEMICONDUCTOR DEVICE

FIELD

[0001] The present disclosure relates to a semiconductor element, and a semiconductor device in which the semiconductor element is mounted.

BACKGROUND

[0002] JP-A-2014-165335 discloses an example of a semiconductor element (a semiconductor device in this patent document). The semiconductor element has a substrate having an element formation surface, a pad terminal provided on the element formation surface, a passivation film covering a portion of the pad terminal and the element formation surface, a Cu redistribution wiring extending out from the pad terminal, an organic film covering the Cu redistribution wiring, and a resin film covering the organic film. The surface of the Cu redistribution wiring includes a roughened surface. The organic film is in contact with the roughened surface. With such a configuration, an anchoring effect for the Cu redistribution wiring is generated in the organic film. Furthermore, because the affinity between the organic film and the resin film is relatively high, the adhesion of the resin film to the organic film is enhanced. This suppresses separation of the resin film from the Cu redistribution wiring.

[0003] The semiconductor element disclosed in JP-A-2014-165335 further includes a terminal electrically conducting to the Cu redistribution wiring, and a bonding layer on the terminal. In the semiconductor element, electromigration due to the current density in the terminal affects the bonding layer. If the degree of electromigration is relatively large, there is a risk that cracks will occur in the bonding layer.

DRAWINGS

[0004] FIG. **1** is a plan view of a semiconductor element according to a first embodiment of the present disclosure.

[0005] FIG. **2** is a partial enlarged view of FIG. **1**, in which a bonding layer is transparent.

[0006] FIG. **3** is a sectional view taken along line III-III in FIG. **2**.

[0007] FIG. **4** is a sectional view taken along line IV-IV in FIG. **2**.

[0008] FIG. **5** is a graph for describing the effects of the semiconductor element shown in FIG. **1**.

[0009] FIG. **6** is a sectional view of a semiconductor device in which the semiconductor element shown in FIG. **1** is mounted.

[0010] FIG. **7** is a partial enlarged sectional view corresponding to FIG. **3**, showing a semiconductor element according to a second embodiment of the present disclosure.

[0011] FIG. **8** is a graph for describing the effects of the semiconductor element shown in FIG. **7**.

[0012] FIG. **9** is a partial enlarged sectional view corresponding to FIG. **3**, showing a semiconductor element according to a third embodiment of the present disclosure.

[0013] FIG. **10** is a partial enlarged plan view corresponding to FIG. **2**, showing a semiconductor element according to a fourth embodiment of the present disclosure.

[0014] FIG. **11** is a sectional view taken along line XI-XI in FIG. **10**.

[0015] FIG. **12** is a sectional view of a semiconductor element according to a fifth embodiment of the present disclosure.

EMBODIMENTS

[0016] The details of the present disclosure will be described based on the accompanying drawings.

First Embodiment

[0017] A semiconductor element A**10** according to a first embodiment of the present disclosure will be described based on FIGS. **1** to **5**. The semiconductor element A**10** is, for example, an LSI (Large Scale Integration) referred to as a Wafer Level-Chip Size Package (WL-CSP). The semiconductor element A**10** includes a main body **11**, a plurality of electrodes **12**, a passivation film **20**, a first protective film **31**, a second protective film **32**, a plurality of redistribution wirings **40**, a plurality of terminals **50**, and a plurality of bonding layers **60**. For the convenience of understanding, the bonding layer **60** is transparent in FIG. **2**.

[0018] In the description of the semiconductor element A**10** and a semiconductor device B that will be described later, the direction that is normal to the obverse surface **11**A of the main body **11**, described later, is referred to as the "first direction z". A direction orthogonal to the first direction z is referred to as the "second direction x". The direction orthogonal to each of the first direction z and the second direction z is referred to as the "third direction y".

[0019] As shown in FIGS. **3** and **4**, the main body **11** includes a semiconductor substrate **111** and a semiconductor layer **112** located on one side of the semiconductor substrate **111** in the first direction z. The main body **11** has an obverse surface **11**A facing the above-mentioned one side in the first direction z. The semiconductor layer **112** includes the obverse surface **11**A. The semiconductor substrate **111** is made from a silicon wafer, for example. Various semiconductor circuits, such as transistors and diodes, are configured on or near the obverse surface **11**A of the semiconductor layer **112**.

[0020] As shown in FIGS. **3** and **4**, the plurality of electrodes **12** are located on the above-mentioned one side of the main body **11** in the first direction z. The electrodes **12** are in contact with the obverse surface **11**A of the main body **11**. The electrodes **12** electrically conduct to various semiconductor circuits configured on the semiconductor layer **112**. The electrodes **12** contain aluminum (Al), for example. The configurations, such as the arrangement and shapes, of the electrodes **12** in the present disclosure are merely examples. As shown in FIG. **1**, the plurality of electrodes **12** include a first electrode **12**A, a second electrode **12**B, and a third electrode **12**C. Each of the first electrode **12**A, the second electrode **12**B, and the third electrode **12**C extends in the third direction y. The first electrode **12**A, the second electrode **12**B, and the third electrode **12**C are arranged side by side along the second direction x. The second electrode **12**B is located next to the first electrode **12**A in the second direction x. The third electrode **12**C is located opposite to the first electrode **12**A across the second electrode **12**B.

[0021] As shown in FIGS. **3** and **4**, the passivation film **20** covers the obverse surface **11**A of the main body **11** and a portion of each electrode **12**. The passivation film **20** is a thin film containing silicon dioxide (SiO_2) or silicon nitride (Si_3N_4), or a laminate of such thin films.

[0022] As shown in FIGS. **2** to **4**, the passivation film **20** has a plurality of openings **21**. The openings **21** penetrate in the first direction z. Each of the first electrode **12**A, the second electrode **12**B, and the third electrode **12**C is exposed from one of the openings **21**.

[0023] As shown in FIGS. **3** and **4**, the second protective film **32** is located between the main body **11** and the first protective film **31** in the first direction z. The second protective film **32** covers a portion of each electrode **12**, and the passivation film **20**. The second protective film **32** is an insulator containing an organic compound. The second protective film **32** is made of a material containing polyimide. The second protective film **32** includes portions located between the passivation film **20** and the plurality of redistribution wirings **40**. As shown in FIG. **3**, the second protective film **32** has second openings **321** penetrating in the first direction z. As viewed in the first direction z, each second opening **321** overlaps with one of the openings **21** of the passivation film **20**. In the semiconductor element A**10**, each second opening **321** exposes one of the electrodes **12**. In the semiconductor element A**10**, one of the second openings **321** exposes the first electrode **12**A.

[0024] As shown in FIGS. **3** and **4**, the redistribution wirings **40** are located between the electrodes **12** and the terminals **50** in the first direction z. Each of the redistribution wirings **40** electrically conducts to one of the electrodes **12**. Each of the redistribution wirings **40** includes a barrier layer that is in contact with at least one of the electrodes **12** and the second protective film **32**, a seed layer deposited on the barrier layer, and a plating layer deposited on the seed layer. The barrier layer contains titanium (Ti). Each of the seed layer and the plating layer contains copper (Cu).

[0025] As shown in FIGS. **3** and **4**, at least one of the redistribution wirings **40** has a main portion **41**, a contact portion **42**, and a dented portion **43**. The contact portion **42** is electrically connected to one of the electrodes **12**. In the semiconductor element A**10**, the contact portion **42** is electrically connected to the first electrode **12**A. The contact portion **42** is in contact with the second protective film **32** and housed in a second opening **321** of the second protective film **32**. As viewed in the first direction z, the entirety of the contact portion **42** overlaps with one of the openings **21** of the passivation film **20**. The main portion **41** is located opposite to the electrodes **12** across the contact portion **42** in the first direction z. The contact portion **42** is connected to the main portion **41**. The main portion **41** is sandwiched between the first protective film **31** and the second protective film **32**. The dented portion **43** is recessed in the first direction z from the side of the main portion **41** that faces the first protective film **31** in the first direction z. As viewed in the first direction z, the dented portion **43** overlaps with the contact portion **42**. The first protective film **31** is received in the dented portion **43**.

[0026] As shown in FIGS. **3** and **4**, the first protective film **31** covers the second protective film **32** and the redistribution wirings **40**. The first protective film **31** is an insulator containing an organic compound. The first protective film **31** is made of a material containing polyimide. In the semiconductor element A**10**, the composition of the first protective film **31** is the same as that of the second protective film **32**. The first protective film **31** is in contact with the terminals **50**. The dimension in the first direction z of the first protective film **31** is greater than the dimension in the first direction z of the second protective film **32**. The first protective film **31** has first openings **311** penetrating in the first direction z. Each first opening **311** exposes the main portion **41** of one of the redistribution wirings **40**. A portion of one of the terminals **50** is housed in each first opening **311**.

[0027] As shown in FIGS. **3** and **4**, each of the terminals **50** is located opposite to the main body **11** with respect to the electrodes **12** and the redistribution wirings **40** in the first direction z. Each of the terminals **50** is electrically connected to the main portion **41** of one of the redistribution wirings **40**. Thus, each of the terminals **50** electrically conducts to one of the electrodes **12**. The terminals **50** contain copper. As shown in FIG. **1**, in the semiconductor element A**10**, the plurality of terminals **50** are arranged in a grid pattern as viewed in the first direction Z.

[0028] As shown in FIGS. **3** and **4**, each of the terminals **50** has a first portion **51** and a second portion **52**. The first portion **51** is housed in a first opening **311** of the first protective film **31**. The second portion **52** is connected to the first portion **51** and protrudes from the first opening **311**. As shown in FIG. **2**, as viewed in the first direction z, the second portion **52** is located outside the first opening **311**.

[0029] As shown in FIGS. **3** and **4**, the first portion **51** has a first surface **511** and a first peripheral surface **512**. The first surface **511** faces the side where the electrodes **12** and the redistribution wirings **40** are located in the first direction z. The first surface **511** is electrically connected to the main portion **41** of one of the redistribution wirings **40**. The first peripheral surface **512** surrounds the first surface **511** as viewed in the first direction z. The first peripheral surface **512** is in contact with the first protective film **31**. The first peripheral surface **512** is inclined with respect to each of the second direction x and the third direction y. The cross-sectional area of the first portion **51** in a direction orthogonal to the first direction z decreases from the second portion **52** toward the first surface **511**.

[0030] As shown in FIGS. **3** and **4**, the second portion **52** has a second surface **521** and a second peripheral surface **522**. The second surface **521** faces away from the first surface **511** of the first portion **51** in the first direction z. The second surface **521** is covered with one of the bonding layers **60**. The second peripheral surface **522** surrounds the second surface **521** as viewed in the first direction z. The second peripheral surface **522** is exposed to the outside of the semiconductor element A**10**. The second peripheral surface **522** is inclined with respect to each of the second direction x and the third direction y. The cross-sectional area of the second portion **52** in a direction orthogonal to the first direction z decreases from the second surface **521** toward the first portion **51**. As viewed in the first direction z, the entirety of the first peripheral surface **512** of the first portion **51** is located inwardly of the second peripheral surface **522**.

[0031] As shown in FIGS. **3** and **4**, in each of the second direction x and the third direction y, the dimension D1 of the periphery of the first surface **511** of the first portion **51** is smaller than the dimension D2 of the periphery of the second surface **521** of the second portion **52**. The dimension D1 is equal to or greater than 40% and equal to or less than 60% of the dimension D2.

[0032] As shown in FIGS. **2** to **4**, each of the terminals **50** has a first recess **53** recessed from the second surface **521** of the second portion **52**. A portion of one of the bonding layers **60** is housed in the first recess **53**.

[0033] As shown in FIG. **3**, each of the bonding layers **60** is located opposite to the first surface **511** across the second surface **521** of one of the terminals **50** in the first direction z. The bonding layers **60** electrically conduct to the terminals **50**, respectively. The bonding layers **60** contain a metal. The bonding layers **60** are solder. Therefore, the composition of the bonding layers **60** includes tin. The melting point of the bonding layers **60** is lower than the melting point of the terminals **50**.

[0034] Next, the semiconductor device B in which the semiconductor element A**10** is mounted will be described based on FIG. **6**.

[0035] As shown in FIG. **6**, the semiconductor device B includes the semiconductor element A**10** and a base **80**. The semiconductor element A**10** is mounted on the base **80**. The base **80** includes a substrate **81** and conductive parts **82**. The substrate **81** is an insulator. The conductive parts **82** contain copper, for example. Each of the terminals **50** of the semiconductor element A**10** is conductively bonded to a conductive part **82** via one of the bonding layers **60**. Thus, the electrodes **12** of the semiconductor elements A**10** electrically conduct to the conductive parts **82**.

[0036] In the semiconductor device B, the base **80** is a wiring board. Alternatively, the base **80** may include only the conductive parts **82** that are leads. The base **80** may include an external terminal located opposite to the conductive parts **82** with respect to the substrate **81** in the first direction z. In the semiconductor device B, the semiconductor element A**10** may be covered with scaling resin such as underfill.

[0037] Next, the effects of the semiconductor element A**10** will be described.

[0038] The semiconductor element A**10** includes the main body **11**, the electrode **12**, the terminal **50**, and the bonding layer **60**. The bonding layer **60** contains a metal. The terminal **50** has the first surface **511** and the second surface **521**. The bonding layer **60** is located opposite to the first surface **511** across the second surface **521**. In the second direction x, the dimension D1 of the periphery of the first surface **511** is equal to or greater than 40% and equal to or less than 60% of the dimension D2 of the periphery of the second surface **521**. Referring now to FIG. **5**, the figure shows the change in the maximum current density at the terminal **50**. The horizontal axis of FIG. **5** shows the ratio [%] of the dimension D1 to the dimension D2. The vertical axis of FIG. **5** shows the maximum current density [A/m^2] at the terminal **50**, where the unit length is along the first direction z. According to FIG. **5**, the inventor has confirmed that the maximum current density at the terminal **50** is relatively low when the ratio of the dimension D1 to the dimension D2 is equal to or greater than 40% and equal to or less than 60%. That is, the present configuration reduces the maximum current density in the terminal **50**, and hence suppresses the occurrence of cracks in the bonding layer **60** due to electromigration. Therefore, according to the present configuration, it is possible to suppress the occurrence of cracks in the bonding layer **60** due to electromigration in the semiconductor element A**10**.

[0039] The semiconductor element A**10** also includes the redistribution wiring **40** and the first protective film **31**. The first protective film **31** has the first opening **311** penetrating in the first direction z and exposing the redistribution wiring **40**. The terminal **50** has the first portion **51** including the first surface **511** and housed in the first opening **311**, and the second portion **52** including the second surface **521** and protruding from the first opening **311**. As viewed in the first direction z, the second portion **52** is located outside the first opening **311**. With such a configuration, the redistribution wiring **40** and the first portion **51** are protected by the first protective film **31**, and excessive changes in the current density at the terminal **50** can be suppressed.

[0040] The first portion **51** has the first peripheral surface **512** surrounding the first surface **511** as viewed in the first direction z. The first peripheral surface **512** is inclined with respect to the second direction x. The cross-sectional area of the first portion **51** in a direction orthogonal to the first direction z decreases from the second portion **52** toward the first surface **511**. With such a configuration, the distribution of current density in the first portion **51** can be made uniform.

[0041] The second portion **52** has the second peripheral surface **522** surrounding the second surface **521** as viewed in the first direction z. The second peripheral surface **522** is inclined with respect to the second direction x. The cross-sectional area of the second portion **52** in a direction orthogonal to the first direction z decreases from the second surface **521** toward the first portion **51**. With such a configuration, the distribution of current density in the second portion **52** can be made uniform.

[0042] As viewed in the first direction z, the entirety of the first peripheral surface **512** of the first portion **51** is located inwardly of the second peripheral surface **522** of the second portion **52**. With such a configuration, the distribution of current density at the interface between the first portion **51** and the second portion **52** can be made uniform.

Second Embodiment

[0043] A semiconductor element A**20** according to a second embodiment of the present disclosure will be described based on FIGS. **7** and **8**. In these figures, the elements that are identical or similar to those of the semiconductor element A**10** described above are denoted by the same reference signs, and the descriptions thereof are omitted. Incidentally, FIG. **7** corresponds to FIG. **3** that shows the semiconductor element A**10**.

[0044] The semiconductor element A**20** differs from the semiconductor element A**10** in the configuration of the terminals **50**.

[0045] As shown in FIG. **7**, in each of the terminals **50**, the dimension t2 of the second portion **52** in the first direction z is greater than the dimension t1 of the first portion **51** in the first direction z. The dimension t2 is equal to or greater than 5% of the dimension D2 in each of the second direction x and the third direction y of the periphery of the second surface **521** of the second portion **52**.

[0046] Next, the effects of the semiconductor element A**20** will be described.

[0047] The semiconductor element A**20** includes the main body **11**, the electrode **12**, the terminal **50**, and the bonding layer **60**. The bonding layer **60** contains a metal. The terminal **50** has the first surface **511** and the second surface **521**. The bonding layer **60** is located opposite to the first surface **511** across the second surface **521**. In the second direction x, the dimension D1 of the periphery of the first surface **511** is equal to or greater than 40% and equal to or less than 60% of the dimension D2 of the periphery of the second surface **521**. With such a configuration, it is possible to suppress the occurrence of cracks in the bonding layer **60** due to electromigration in the semiconductor element A**20** as well. The semiconductor element A**20** has a configuration in common with the semiconductor element A**10**, thereby achieving the same effect as the semiconductor element A**10**.

[0048] In the terminals **50** of the semiconductor element A**20**, the dimension t2 of the second portion **52** in the first direction z is greater than the dimension t1 of the first portion **51** in the first direction z. The dimension t2 is equal to or greater than 5% of the dimension D2 in the second direction x of the periphery of the second surface **521** of the second portion **52**. Referring now to FIG. **8**, the figure shows the change in the maximum current density at the terminal **50**. The horizontal axis of FIG. **8** shows the ratio [%] of the dimension t2 to the dimension D2. The vertical axis of FIG. **8** shows the maximum current density [A/mm^2] at the terminal **50**, where the unit length is along the first direction z. According to FIG. **8**, the inventor has confirmed that the maximum current density at the terminal **50** is relatively low when the ratio of dimension t2 to dimension D2 exceeds 4%. That is, the present configuration reduces the maximum current density in the terminal **50**, and hence suppresses the occurrence of cracks in the bonding layer **60** due to electromigration.

Third Embodiment

[0049] A semiconductor element A**30** according to a third embodiment of the present disclosure will be described based on FIG. **9**. In these figures, the elements that are identical or similar to those of the semiconductor element A**10** described above are denoted by the same reference signs, and the descriptions thereof are omitted. Incidentally, FIG. **9** corresponds to FIG. **3** that shows the semiconductor element A**10**.

[0050] The semiconductor element A**30** differs from the semiconductor element A**20** in that the semiconductor element A**30** further includes intermediate layers **70**.

[0051] As shown in FIG. **9**, each intermediate layer **70** is located between the second surface **521** of one of the terminals **50** and one of the bonding layers **60**. Each intermediate layer **70** is in contact with a second surface **521**, the surface of one of the terminals **50** that defines the first recess **53**, and one of the bonding layers **60**. The intermediate layers **70** contain a metal. The composition of the intermediate layers **70** includes nickel (Ni).

[0052] Next, the effects of the semiconductor element A**30** will be described.

[0053] The semiconductor element A**30** includes the main body **11**, the electrode **12**, the terminal **50**, and the bonding layer **60**. The bonding layer **60** contains a metal. The terminal **50** has the first surface **511** and the second surface **521**. The bonding layer **60** is located opposite to the first surface **511** across the second surface **521**. In the second direction x, the dimension D1 of the periphery of the first surface **511** is equal to or greater than 40% and equal to or less than 60% of the dimension D2 of the periphery of the second surface **521**. With such a configuration, it is possible to suppress the occurrence of cracks in the bonding layer **60** due to electromigration in the semiconductor element A**30** as well. The semiconductor element A**30** has a configuration in common with the semiconductor element A**10**, thereby achieving the same effect as the semiconductor element A**10**.

[0054] The semiconductor element A**30** further includes the intermediate layer **70** located between the second surface **521** of the terminal **50** and the bonding layer **60**. The composition of the intermediate layer **70** includes nickel. The intermediate layer **70** is in contact with each of the second surface **521** and the bonding layer **60**. With such a configuration, when the semiconductor element A**30** is mounted on the base **80** shown in FIG. **6**, eutectic metal is formed by the intermediate layer **70** and the molten bonding layer **60**. The eutectic metal increases the tensile strength of the bonding layer **60**. This effectively suppresses the occurrence of cracks in the bonding layer **60** due to electromigration.

Fourth Embodiment

[0055] A semiconductor element A**40** according to a fourth embodiment of the present disclosure will be described based on FIGS. **10** and **11**. In these figures, the elements that are identical or similar to those of the semiconductor element A**10** described above are denoted by the same reference signs, and the descriptions thereof are omitted. For the convenience of understanding, the bonding layer **60** is transparent in FIG. **10**. Incidentally, FIG. **10** corresponds to FIG. **2** that shows the semiconductor element A**10**.

[0056] The semiconductor element A**40** differs from the semiconductor element A**30** in the configurations of the second protective film **32**, the redistribution wirings **40**, and the terminals **50**.

[0057] As shown in FIGS. **10** and **11**, one of the second openings **321** of the second protective film **32** exposes the second electrode **12**B. As viewed in the first direction z, the second opening **321** overlaps with one of the openings **21** of the passivation film **20** that exposes the second electrode **12**B. The contact portion **42** of one of the redistribution wirings **40** is electrically connected to the second electrode **12**B. The contact portion **42** is housed in the second opening **321**.

[0058] As shown in FIG. **11**, the first portion **51** of each of the terminals **50** is received in the dented portion **43** of one of the redistribution wirings **40**. Each of the terminals **50** has a second recess **54**. The second recess **54** is recessed in the first direction z from the surface of the terminal **50** that defines the first recess **53**. A portion of one of the bonding layers **60** is housed in the second recess **54**. The intermediate layer **70** is in contact with the surface of the terminal **50** that defines the second recess **54**. As shown in FIG. **10**, as viewed in the first direction z, the second recess **54** is located inwardly of the second opening **321** of the second protective film **32**.

[0059] Next, the effects of the semiconductor element A**40** will be described.

[0060] The semiconductor element A**40** includes the main body **11**, the electrode **12**, the terminal **50**, and the bonding layer **60**. The bonding layer **60** contains a metal. The terminal **50** has the first surface **511** and the second surface **521**. The bonding layer **60** is located opposite to the first surface **511** across the second surface **521**. In the second direction x, the dimension D1 of the periphery of the first surface **511** is equal to or greater than 40% and equal to or less than 60% of the dimension D2 of the periphery of the second surface **521**. With such a configuration, it is possible to suppress the occurrence of cracks in the bonding layers **60** due to electromigration in the semiconductor element A**40** as well. The semiconductor element A**40** has a configuration in common with the semiconductor element A**10**, thereby achieving the same effect as the semiconductor element A**10**.

[0061] In the semiconductor element A**40**, each terminal **50** has a second recess **54** recessed in the first direction z from the surface defining the first recess **53**. A portion of a bonding layer **60** is housed in the second recess **54**. Such a configuration enhances the anchoring effect of the bonding layer **60** for the terminal **50**, so that the bonding strength of the bonding layer **60** to the terminal **50** is increased.

[0062] In the semiconductor element A**40**, the first portion **51** of the terminal **50** is received in the dented portion **43** of the redistribution wiring **40**. Such a configuration reduces the electrical resistance at the interface between the redistribution wiring **40** and the terminal **50**.

Fifth Embodiment

[0063] A semiconductor element A**50** according to a fifth embodiment of the present disclosure will be described based on FIG. **12**. In these figures, the elements that are identical or similar to those of the semiconductor element A**10** described above are denoted by the same reference signs, and the descriptions thereof are omitted. Incidentally, FIG. **12** corresponds to FIG. **6** that shows the semiconductor element A**10** constituting the semiconductor device B.

[0064] The semiconductor element A**50** differs from the semiconductor element A**10** in that the semiconductor element A**50** further includes a third protective film **33**.

[0065] As shown in FIG. **12**, the third protective film **33** is located opposite to the electrodes **12** across the main body **11** in the first direction z. The third protective film **33** covers one side of the main body **11** in the first direction z. The third protective film **33** is made of a material containing, for example, an epoxy resin.

[0066] Next, the effects of the semiconductor element A**50** will be described.

[0067] The semiconductor element A**50** includes the main body **11**, the electrode **12**, the terminal **50**, and the bonding layer **60**. The bonding layer **60** contains a metal. The terminal **50** has the first surface **511** and the second surface **521**. The bonding layer **60** is located opposite to the first surface **511** across the second surface **521**. In the second direction x, the dimension D1 of the periphery of the first surface **511** is equal to or greater than 40% and equal to or less than 60% of the dimension D2 of the periphery of the second surface **521**. With such a configuration, it is possible to suppress the occurrence of cracks in the bonding layers **60** due to electromigration in the semiconductor element A**50** as well. The semiconductor element A**50** has a configuration in common with the semiconductor element A**10**, thereby achieving the same effect as the semiconductor element A**10**.

[0068] The semiconductor element A**50** further includes a third protective film **33** located opposite to the electrodes **12** across the main body **11** in the first direction z, and the third protective film **33** covers one side of the main body **11** in the first direction z. With such a configuration, warping of the main body **11** in the first direction z can be reduced during the manufacture of the semiconductor element A**50**.

[0069] The present disclosure is not limited to the above-described embodiments. Various modification in design may be made freely in the specific structure of each part of the present disclosure.

[0070] The present disclosure includes the embodiments described in the following clauses.

Clause 1.

[0071] A semiconductor element (A**10**) comprising:

[0072] a main body (**11**) including a semiconductor layer (**112**);

[0073] an electrode (**12**) located on one side of the main body (**11**) in a first direction and electrically conducting to the semiconductor layer (**112**);

[0074] a terminal (**50**) located opposite to the main body (**11**) with respect to the electrode (**12**) in the first direction and electrically conducting to the electrode (**12**); and

[0075] a bonding layer (**60**) electrically conducting to the terminal (**50**), wherein

[0076] the bonding layer (**60**) contains a metal,

[0077] the terminal (**50**) includes a first surface (**511**) facing a side where the electrode (**12**) is located in the first direction, and a second surface (**521**) facing away from the first surface (**511**) in the first direction,

[0078] the bonding layer (**60**) is located opposite to the first surface (**511**) across the second surface (**521**), and

[0079] in a second direction orthogonal to the first direction, a dimension (D1) of a periphery of the first surface (**511**) is equal to or greater than 40% and equal to or less than 60% of a dimension (D2) of a periphery of the second surface (**521**).

Clause 2.

[0080] The semiconductor element (A**10**) according to clause 1, wherein composition of the bonding layer (**60**) includes tin.

Clause 3.

[0081] The semiconductor element (A**10**) according to clause 2,

[0082] further comprising a redistribution wiring (**40**) located between the electrode (**12**) and the terminal (**50**) in the first direction, wherein the redistribution wiring (**40**) electrically conducts to the electrode (**12**) and the terminal (**50**).

Clause 4.

[0083] The semiconductor element (A**10**) according to clause 3, wherein each of the first surface (**511**) and the electrode (**12**) is electrically connected to the redistribution wiring (**40**).

Clause 5.

[0084] The semiconductor element (A**10**) according to clause 4,

[0085] further comprising a first protective film (**31**) covering the redistribution wiring (**40**), wherein

[0086] the first protective film (**31**) is formed with a first opening (**311**) penetrating in the first direction and exposing the redistribution wiring (**40**), and

[0087] a portion of the terminal (**50**) is housed in the first opening (**311**).

Clause 6.

[0088] The semiconductor element (A**10**) according to clause 5, wherein the terminal (**50**) includes a first portion (**51**) including the first surface (**511**) and housed in the first opening (**311**), and a second portion (**52**) including the second surface (**521**) and protruding from the first opening (**311**), and

[0089] the second portion (**52**) is located outside the first opening (**311**) as viewed in the first direction.

Clause 7.

[0090] The semiconductor element (A**10**) according to clause 6, wherein the first portion (**51**) includes a first peripheral surface (**512**) surrounding the first surface (**511**) as viewed in the first direction, and

[0091] the first peripheral surface (**512**) is in contact with the first protective film (**31**).

Clause 8.

[0092] The semiconductor element (A**10**) according to clause 7, wherein the first peripheral surface (**512**) is inclined with respect to the second direction, and

[0093] a cross-sectional area of the first portion (**51**) in a direction orthogonal to the first direction decreases from the second portion (**52**) toward the first surface (**511**).

Clause 9.

[0094] The semiconductor element (A**10**) according to clause 8, wherein the second portion (**52**) includes a second peripheral surface (**522**) surrounding the second surface (**521**) as viewed in the first direction, and

[0095] the first peripheral surface (**512**) is entirely located inwardly of the second peripheral surface (**522**) as viewed in the first direction.

Clause 10.

[0096] The semiconductor element (A**10**) according to clause 9, wherein the second peripheral surface (**522**) is inclined with respect to the second direction, and

[0097] a cross-sectional area of the second portion (**52**) in a direction orthogonal to the first direction decreases from the second surface (**521**) toward the first portion (**51**).

Clause 11.

[0098] The semiconductor element (A**20**) according to clause 10, wherein a dimension (t2) of the second portion (**52**) in the first direction is greater than a dimension (t1) of the first portion (**51**) in the first direction.

Clause 12.

[0099] The semiconductor element (A**20**) according to clause 11, wherein the dimension (t2) of the second portion (**52**) in the first direction is equal to or greater than 5% of the dimension (D2) of the periphery of the second surface (**521**) in the second direction.

Clause 13.

[0100] The semiconductor element (A**30**) according to clause 12, further comprising an intermediate layer (**70**) located between the second surface (**521**) and the bonding layer (**60**), wherein

[0101] the intermediate layer (**70**) contains a metal, and

[0102] the intermediate layer (**70**) is in contact with each of the second surface (**521**) and the bonding layer (**60**).

Clause 14.

[0103] The semiconductor element (A**30**) according to clause 13, wherein composition of the intermediate layer (**70**) includes nickel.

Clause 15.

[0104] The semiconductor element (A**10**) according to any one of clauses 5 to 14, wherein the terminal (**50**) includes a first recess (**53**) recessed from the second surface (**521**), and

[0105] a portion of the bonding layer (**60**) is housed in the first recess (**53**).

Clause 16.

[0106] The semiconductor element (A**10**) according to clause 15, further comprising a second protective film (**32**) located between the main body (**11**) and the redistribution wiring (**40**) in the first direction, wherein

[0107] the second protective film (**32**) includes a second opening (**321**) penetrating in the first direction and exposing the electrode (**12**), and

[0108] a portion of the redistribution wiring (**40**) is housed in the second opening (**321**).

Clause 17.

[0109] The semiconductor element (A**10**) according to clause 16, wherein the first recess (**53**) overlaps with the second opening (**321**) as viewed in the first direction.

Clause 18.

[0110] The semiconductor element (A**40**) according to clause 17, wherein the terminal (**50**) includes a second recess (**54**) recessed in the first direction from a surface of the terminal (**50**) that defines the first recess (**53**), and

[0111] a portion of the bonding layer (**60**) is housed in the second recess (**54**).

Clause 19.

[0112] A semiconductor device (B) comprising:

[0113] the semiconductor element (A**10**) as set forth in clause 15, and

[0114] a base (**80**) that includes a conductive part (**82**), wherein

[0115] the semiconductor element (A**10**) is mounted on the base (**80**), and

[0116] the terminal (**50**) is conductively bonded to the conductive part (**82**) via the bonding layer (**60**).

Clause 20.

[0117] The semiconductor element (A**10**) according to clause 16, wherein each of the first protective film (**31**) and the second protective film (**32**) contains polyimide.

Clause 21.

[0118] The semiconductor element (A**10**) according to clause 16, wherein a dimension of the first protective film (**31**) in the first direction is greater than the second dimension (**32**) of the second protective film.

Clause 22.

[0119] The semiconductor element (A**10**) according to clause 16, wherein the redistribution wiring (**40**) includes a main portion (**41**) located between the first protective film (**31**) and the second protective film (**32**), and a contact portion (**42**) connected to the main portion (**41**),

[0120] the electrode (**12**) is electrically connected to the contact portion (**42**) and in contact with the second protective film (**32**), and

[0121] the first surface (**511**) is electrically connected to the main portion (**41**).

Clause 23.

[0122] The semiconductor element (A**10**) according to clause 22, wherein the contact portion (**42**) is housed in the second opening (**321**).

Clause 24.

[0123] The semiconductor element (A**10**) according to clause 23, wherein the redistribution wiring (**40**) includes a dented portion (**43**) recessed from the main portion (**41**) in a first direction,

[0124] the dented portion (**43**) overlaps with the contact portion (**42**) as viewed in the first direction, and

[0125] the first protective film (**31**) is received in the dented portion (**43**).

Clause 25.

[0126] The semiconductor element (A**40**) according to clause 18, wherein the redistribution wiring (**40**) includes a dented portion (**43**) recessed from one side in a first direction, and

[0127] a part of the first portion (**51**) is received in the dented portion (**43**).

Clause 26.

[0128] The semiconductor element (A**40**) according to clause 18, wherein the second recess (**54**) is located inwardly of the second opening (**321**) as viewed in the first direction.

Clause 27.

[0129] The semiconductor element (A**50**) according to clause 16, further comprising a third protective film (**33**) located opposite to the electrode (**12**) across the main body (**11**),

[0130] wherein the third protective film (**33**) covers one side of the main body (**11**) in the first direction.

REFERENCE NUMERALS

A10 to A50: Semiconductor element	B: Semiconductor device
11: Main body	11A: Obverse surface
111: Semiconductor substrate	112: Semiconductor layer
12: Electrode	12A, 12B, 12C: First electrode, Second electrode, Third electrode
20: Passivation film	21: Opening
31: First protective film	311: First opening
32: Second protective film	321: Second opening
33: Third protective film	40: Redistribution wiring
41: Main portion	42: Contact portion
43: Dented portion	50: Terminal
51: First portion	511: First surface

-continued

REFERENCE NUMERALS

512: First peripheral surface	52: Second portion
521: Second surface	522: Second peripheral surface
53, 54: First recess, Second recess	60: Bonding layer
70: Intermediate layer	80: Base
81: Substrate	82: Conductive part
z, x, y: First direction, Second direction, Third direction	

1. A semiconductor element comprising:

a main body including a semiconductor layer;

an electrode located on one side of the main body in a first direction and electrically conducting to the semiconductor layer;

a terminal located opposite to the main body with respect to the electrode in the first direction and electrically conducting to the electrode; and

a bonding layer electrically conducting to the terminal, wherein

the bonding layer contains a metal,

the terminal includes a first surface facing a side where the electrode is located in the first direction, and a second surface facing away from the first surface in the first direction,

the bonding layer is located opposite to the first surface across the second surface, and

in a second direction orthogonal to the first direction, a dimension of a periphery of the first surface is equal to or greater than 40% and equal to or less than 60% of a dimension of a periphery of the second surface.

2. The semiconductor element according to claim **1**, wherein composition of the bonding layer includes tin.

3. The semiconductor element according to claim **2**, further comprising a redistribution wiring located between the electrode and the terminal in the first direction,

wherein the redistribution wiring electrically conducts to the electrode and the terminal.

4. The semiconductor element according to claim **3**, wherein each of the first surface and the electrode is electrically connected to the redistribution wiring.

5. The semiconductor element according to claim **4**, further comprising a first protective film covering the redistribution wiring, wherein

the first protective film is formed with a first opening penetrating in the first direction and exposing the redistribution wiring, and

a portion of the terminal is housed in the first opening.

6. The semiconductor element according to claim **5**, wherein the terminal includes a first portion including the first surface and housed in the first opening, and a second portion including the second surface and protruding from the first opening, and

the second portion is located outside the first opening as viewed in the first direction.

7. The semiconductor element according to claim **6**, wherein the first portion includes a first peripheral surface surrounding the first surface as viewed in the first direction, and

the first peripheral surface is in contact with the first protective film.

8. The semiconductor element according to claim **7**, wherein the first peripheral surface is inclined with respect to the second direction, and

a cross-sectional area of the first portion in a direction orthogonal to the first direction decreases from the second portion toward the first surface.

9. The semiconductor element according to claim **8**, wherein the second portion includes a second peripheral surface surrounding the second surface as viewed in the first direction, and

the first peripheral surface is entirely located inwardly of the second peripheral surface as viewed in the first direction.

10. The semiconductor element according to claim **9**, wherein the second peripheral surface is inclined with respect to the second direction, and

a cross-sectional area of the second portion in a direction orthogonal to the first direction decreases from the second surface toward the first portion.

11. The semiconductor element according to claim **10**, wherein a dimension of the second portion in the first direction is greater than a dimension of the first portion in the first direction.

12. The semiconductor element according to claim **11**, wherein the dimension of the second portion in the first direction is equal to or greater than 5% of the dimension of the periphery of the second surface in the second direction.

13. The semiconductor element according to claim **12**, further comprising an intermediate layer located between the second surface and the bonding layer, wherein

the intermediate layer contains a metal, and

the intermediate layer is in contact with each of the second surface and the bonding layer.

14. The semiconductor element according to claim **13**, wherein composition of the intermediate layer includes nickel.

15. The semiconductor element according to claim **5**, wherein the terminal includes a first recess recessed from the second surface, and

a portion of the bonding layer is housed in the first recess.

16. The semiconductor element according to claim **15**, further comprising a second protective film located between the main body and the redistribution wiring in the first direction, wherein

the second protective film includes a second opening penetrating in the first direction and exposing the electrode, and

a portion of the redistribution wiring is housed in the second opening.

17. The semiconductor element according to claim **16**, wherein the first recess overlaps with the second opening as viewed in the first direction.

18. The semiconductor element according to claim **17**, wherein the terminal includes a second recess recessed in the first direction from a surface of the terminal that defines the first recess, and

a portion of the bonding layer is housed in the second recess.

19. A semiconductor device comprising:

the semiconductor element as set forth in claim **15**, and

a base that includes a conductive part, wherein

the semiconductor element is mounted on the base, and

the terminal is conductively bonded to the conductive part via the bonding layer.

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